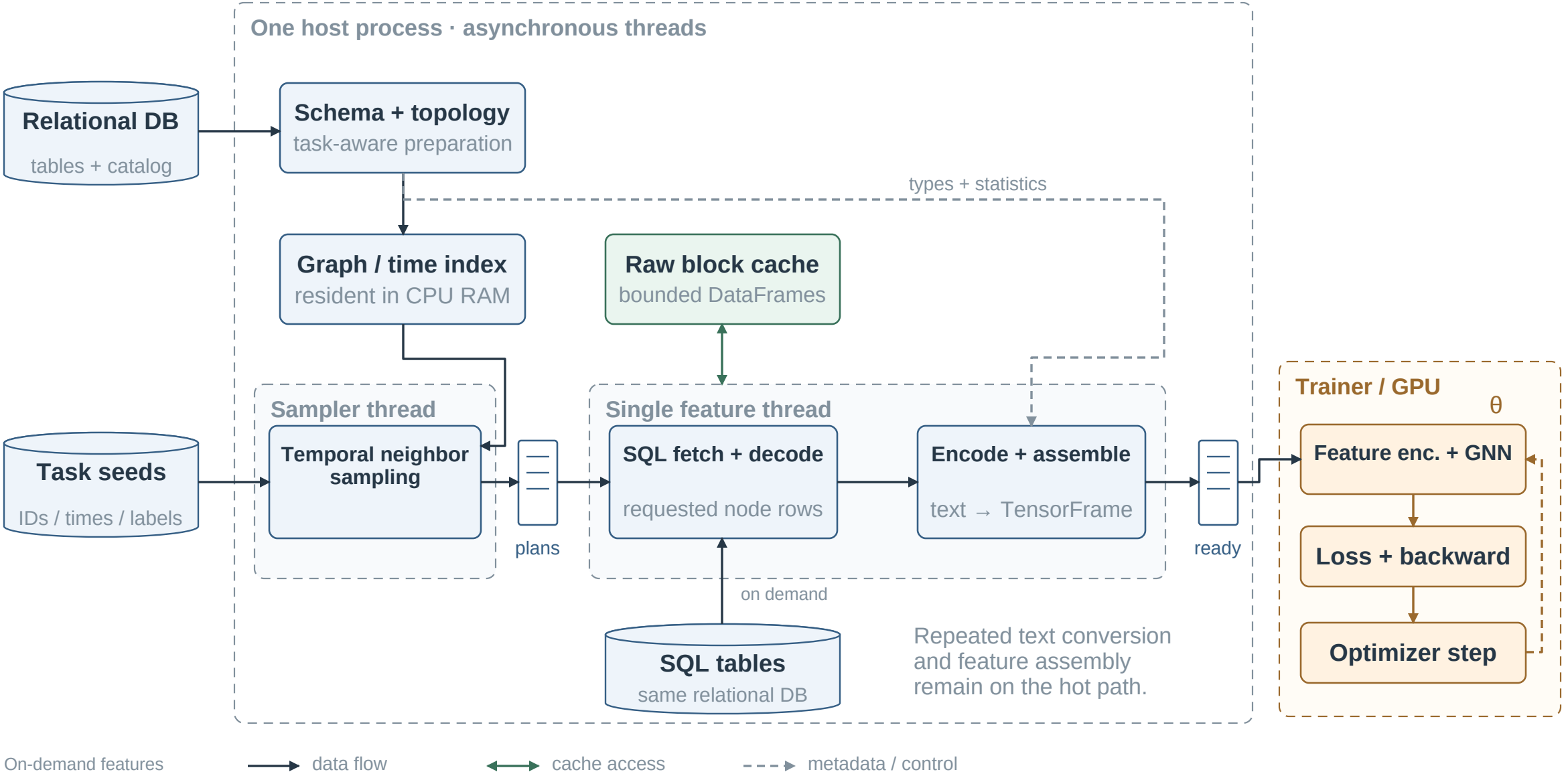
One host process · asynchronous threads
Relational DB
tables + catalog
Schema + topology
task-aware preparation
types + statistics
Graph / time index
resident in CPU RAM
Raw block cache
bounded DataFrames
Sampler thread
Single feature thread
Trainer / GPU
θ
Task seeds
IDs / times / labels
Temporal neighbor sampling
plans
SQL fetch + decode
requested node rows
Encode + assemble
text → TensorFrame
ready
Feature enc. + GNN
Loss + backward
Optimizer step
on demand
SQL tables
same relational DB
Repeated text conversion and feature assembly remain on the hot path.
On-demand features
data flow
cache access
metadata / control